

Over the 17 years tested from 1988 to 2004, portfolios of 30 stocks with the best combination of *earnings yield* and *return on capital* would have returned 30.8 percent per year. To put this in context, wrote Greenblatt, investing at 30.8 percent per year for 17 years would have turned \$11,000 into well over \$1 million. Over the same period, the market returned just 12.4 percent per year, which would have turned the same \$11,000 into just \$79,000. Further, Greenblatt found that his portfolios generated those returns while taking much less risk than the overall market. The smallest company in Greenblatt's 3,500 stock universe had a market capitalization of just \$50 million. If he limited the universe to the largest 2,500 stocks—this time the smallest company in the universe would have a market capitalization of \$200 million—the Magic Formula's average annual return dropped to a still very impressive 23.7 percent, almost double the market's return. In a universe limited to just the largest 1,000 stocks—the smallest company would have a market capitalization in excess of \$1 billion—the Magic Formula still generated an average annual return of 22.9 percent. The three computer simulations of Greenblatt's quantified and simplified version of Buffett's *wonderful company at a fair price* investment strategy suggested that it would have substantially beaten the market, and done so with less risk. Greenblatt was so impressed with the results that he called the simple criteria the "Magic Formula," and wrote about it in a 2006 book called *The Little Book That Beats The Market*.¹¹ Greenblatt has now built his mechanical Buffett. With business partner Robert Goldstein he founded an investment firm, Gotham Asset Management, to systematically implement the Magic Formula in a variety of different investment vehicles.

Many investors, when exposed to the Magic Formula, seem skeptical that such simple criteria could reliably select stocks that beat the market on average, and by such a wide margin. The criticism clusters around two main objections. First, the critics claim the Magic Formula will not work as it did in the test period because it is a product of data mining. Data mining is the repeated examination of a data set to uncover a relationship that exists only coincidentally, and is therefore unlikely to persist outside the data set. The critics imply that Greenblatt tested many different factors and combinations of factors until he found one that beat the market. He then retroactively fitted a plausible-sounding explanation to those factors, and the Magic Formula was the result. The second criticism is that the Magic Formula does select *stocks* that appear to beat the market, but an *investor* could not use it to beat the market because the stocks selected are too small and illiquid. Curious, with Wes Gray, a 2010 Ph.D. graduate from the University of Chicago's Booth School of Business finance program, I analyzed Greenblatt's Magic Formula for our book *Quantitative Value* (2012).¹² We tested the Magic Formula to the academic gold standard,